

STATISTICAL MACHINE LEARNING (WS2025/26)

HOMEWORK: ML ESTIMATE AND PLUGIN BAYES CLASSIFIER

Assignment (4 points) Consider a multi-class classification problem with the following setup. The input features lie in $\mathcal{X} = \mathbb{R}^n$, the hidden state is from $\mathcal{Y} = \{0, 1, \dots, Y - 1\}$, representing the class label, and the loss function $\ell: \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+$. You are provided with the training set $\mathcal{T}^m = (\mathbf{x}^i, y^i) \in \mathcal{X} \times \mathcal{Y} \mid i = 1, \dots, m$. The task involves:

a) Modeling the training data. Fit the training data using mixture of multivariate normal distributions

$$p_{\theta}(\mathbf{x}, y) = \frac{p(y)}{(2\pi)^{n/2} |\mathbf{V}|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_y)^T \mathbf{V}^{-1} (\mathbf{x} - \boldsymbol{\mu}_y) \right].$$

whose parameters θ involve i) the prior probabilities $p(y)$, $y \in \mathcal{Y}$, ii) the mean vectors $\boldsymbol{\mu}_y \in \mathbb{R}^n$, $y \in \mathcal{Y}$, and iii) a common covariance matrix $\mathbf{V} \in \mathbb{R}^{n \times n}$. The parameters of the mixture model should be estimated using the Maximum-Likelihood estimation method.

b) Constructing a classifier. Use the estimated model $p_{\theta_m}(\mathbf{x}, y)$ to create a plug-in Bayes classifier:

$$h(\mathbf{x}) \in \underset{\hat{y} \in \mathcal{Y}}{\text{Argmin}} \sum_{y \in \mathcal{Y}} p_{\theta_m}(\mathbf{x}, y) \ell(y, \hat{y})$$

This classifier should not only provide predictions for the class labels but also quantify the uncertainty associated with these predictions by computing the conditional risk:

$$r(\mathbf{x}) = \sum_{y \in \mathcal{Y}} p_{\theta_m}(y \mid \mathbf{x}) \ell(y, h(\mathbf{x})).$$